# Counting and Probability

R Pruim

2020-03-15

## Contents

# Preface

Placeholder

# 1 Some Counting Problems

Placeholder

**Answers/Solutions**

# 2 Pigeonhole Principle

Placeholder

# 3 Division Principle

Placeholder

## 3.1 A problem about cookies

## 3.2 The Division Principle

## 3.3 More Counting Problems

# 4 Counting Problems

Placeholder

## 4.1 Donut problems

## 4.2 More Problems

# 5 Probability

Placeholder

## 5.1 Notation Note

## 5.2 Probability Axioms

## 5.3 Some Additional Probability Rules

## 5.4 Exercises

# 6 Conditional Probability

Placeholder

## 6.1 Exercises

## 6.2 Product Rule

## 6.3 Independence

## 6.4 Exercises

# 7 Probability Trees and Bayes

Placeholder

## 7.1 Bob's Beautiful Boxes

## 7.2 Breast cancer screening

## 7.3 Disease Testing

## 7.4 Random Primality Testing

## 7.5 Hashing

# 8 Random Variables

Placeholder

## 8.1 Describing Random Variables

## 8.2 Probability Tables

### First example

### Creating a probability table

## 8.3 Probability Functions

## 8.4 Binomial Distributions

# 9 Expected Value

Placeholder

## 9.1 An Example: GPA

## 9.2 Generalizing